Better Data, Smarter Detection: How Training Data Shapes AI-Generated Image Detectors

ABSTRACT

AI-image generation models have rapidly advanced, making it easier than ever to make hard to detect synthetic images. Synthetic images can be used for harmful purposes, such as to spread misinformation or copyright infringement. Many AI image detectors have been developed to address this issue but continue to struggle on difficult modern AI-images. For our research, we examined four modern detectors. Our experiments showed that no single model was clearly the best in terms of AI-image detection accuracy. We found that increasing the diversity and amount of training data used for detectors was an easy and effective way of improving accuracy, increasing by up to 11% on AI-image datasets. However, increasing the amount of training data can have diminishing returns, sometimes worsening performance overall. Our code is avaliable here: https://github.com/JohnTranTh/AI-Image-Detection-Research.

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1 INTRODUCTION

Advancements in generative models, such as with GANs and diffusion models, have shown great ability to produce highly realistic and high quality images that have become increasingly difficult to distinguish from real images. The availability of these methods can have positive effects, such as assisting artists and designers while lowering the barrier to entry for people in general. However, they also have large risks and concerns, such as spreading misinformation, creating deepfakes of others, and infringing copyright protections.

To address these issues, numerous Artificial Intelligence Generated Image (AIGI) Detectors have been created to help distinguish real images from synthetic ones. However, many existing AIGI detectors perform poorly on modern benchmark datasets, especially in-the-wild datasets that source images from the internet such as Chameleon, which contain heavily post-processed images. In this paper, we primarily examine four modern detectors: AIDE[8], SAFE[3], Aligned Forensics[6], and ZeroFake[7]. The first three models train binary classifiers to distinguish between real and synthetic images and report high accuracy across a wide variety of generation models. ZeroFake uses a zero-training method of detecting AI-images.

In order to improve the performance of these models, we perform a variety of experiements on these models and study the effects of changing the architecture of the models versus retraining with new data and compare the results. We propose two methods of improving these models performance. For AIDE, we introduce multi-scale patches, where additional patches of varying sizes are used to influence the model's decision making. For SAFE and Aligned Forensics, we retrain the models with new datasets, along with experiments on various training settings modifications to improve accuracy. For ZeroFake, we use the openly avaliable code to compare the results of a no-training detector to our new methods.

2 RELATED WORKS

AIGI Detectors: AIDE[8] detects inconsistencies in images by combining two types of features: high-level content and low-level details. It uses two main parts: a Semantic Feature Extractor, which looks for large-scale content issues in the image using CLIP-ConvNeXt embeddings, and a Patchwise Feature Extractor which analyzes image patches based on their frequency and uses a lightweight CNN to find detailed noise or artifact patterns.

SAFE[3] improves deepfake detection by addressing two main challenges: loss of details from aggressive downsampling and overfitting to superficial cues like color or layout. Instead of changing the model itself, SAFE tweaks how the data is prepared before feeding it to the model. It replaces resizing with random cropping to keep more details, uses techniques like Color-Jitter and RandomRotation to reduce over-reliance on color or layout patterns, and adds patch-level random masking to help the model focus on specific areas where fake pixels usually appear.

Aligned Forensics[6] aims to have detectors focus on the unique patterns left by generative models and not on superficial cues like image content or quality. They propose a better way of constructing training datasets by reconstructing real images an latent diffusion model's autoencoder without applying any noise-reducing steps, creating near identical reconstructions to the originals except with subtle differences left by the LDM. This apporach alligns the real and fake images, forcing the detectors to focus on these small artifacts rather than irrelevant features.

ZeroFake[7] takes a zero-training approach to AIGI detection. It uses a technique called DDIM inversion, which involves adding noise to an image and then reconstructing it. Fake images behave differently from real ones during this process, so by comparing the reconstructed image to the original, ZeroFake can determine if the image is real or fake.

AI-Generated Image Datasets Chameleon[8] has three key features: Magnitude: It includes about 26,000 test images, making it large and robust. Diversity: The dataset covers a wide range of real-world scenarios, unlike other datasets that focus on just a few

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categories. Resolution: With images ranging from 720P to 4K, the high quality presents extra challenges for models, requiring more detailed analysis. In short, Chameleon provides a tougher and more practical benchmark for improving AI image detection.

AIGIBench [4] aims to create a comprehensive benchmark dataset of AI-images for testing robustness and generalization ability of AI-generated image detectors. Evaluation and training datasets are sourced from this benchmark.

Improving Training Data Selection Guillaro et al.'s [1] paper demonstrates that biases in the training data can negatively impact detector performance. These may include features like compression, resolution, or semantic content. To improve training datasets, the paper proposes an image reconstruction process similar to Aligned Forensics where detectors are trained on closely aligned real and fake images to reduce biases and focus on artifacts introduced by generative models.

Konstantinidou et al.'s [2] paper examines the effects of different training datasets on AIGI detector performance on in-the-wild datasets of AI images.The paper finds that many modern detectors tend to improve when trained on datasets consisting of AI images sourced from in-the-wild sources, such as social media websites, but performance can degrade when training datasets become to large.

Park and Owens' [5] paper researches the effects of increasing the diversity of the training dataset by increasing the number of image generators a training dataset sources from. Accuracy and generalization typically improves as the diversity and size of the dataset increases, but there are diminishing returns as the number of datasets/images increases.

3 METHODOLOGY

For AIDE [8], the main area of interest in the model for modification is the Patchwise Feature Extraction method. The model examines low-level patch statistics between AI-generated images and real images. The model uses a Discrete Cosine Transform (DCT) score module to find the patches with the highest and lowest frequencies. For an image, it is divided into multiple patches with a fixed size (32 x 32 pixels in the case of AIDE). DCT is then applied to each patch to get their results in the frequency domain and find the highest and lowest frequency patches. These patches are then resized to size 256 x 256 pixels, input into an SRM filter to obtain a noise pattern, then further input into two ResNet-50 networks to obtain the final feature map.

Our modification is to add multi-scale patch evaluation into this method. Instead of only examining patches of a fixed size, the image splitting method can be run multiple times for multiple different patch sizes. In our experiments, we test a combination of 16 x 16 patches and 32 x 32 patches, along with a combination of 16, 32, and 64 x 64 patches. Afterwards, the highest and lowest frequency patches for each size is picked and resized and inputted as normal. We find that this method of changing the model fundamentals has mixed results when testing on various AI-image datasets, therefore we look to the more efficient method of changing training data to improve detectors' performance.

For SAFE[3], the model is trained on images from ForenSynths. The training set contains 20 classes, each with 18,000 synthetic images generated from ProGAN and an equal amount of real images from the LSUN dataset. The SAFE model is trained with 4 classes - car, cat, chair, and horse.

Our modification is to increase the size and diversity of the training dataset with images from other generation methods. We first test training with the original dataset paired with 20k Stable Diffusion images (10k real, 10k fake) from the AIGIBench dataset. We also test training the detector with those two datasets combined with 10k real and 10k fake images used to train the original Aligned Forensics model. The fake images are reconstructed from the real images using the method proposed in the original research paper. All training was completed using the original training settings from the research paper. Experiments were performed on an NVIDIA RTX 4070 Super 12GB. We find that increasing the size and diversity of the training dataset generally improves detector performance when using the ProGAN + Stable Diffusion variation, but more training data does not always equate to better performance. When a third dataset, the Aligned Forensics training dataset, is added, performance generally decreases.

4 EXPERIMENTAL RESULTS

Collection of experimental results from testing various ways of improving detector performance. Tables 1 and 2 are of the most interest for this paper.

Tested Dataset	AIDE Model Accuracy	Aligned Forensics Model	SAFE Model
Chameleon	62.89%	**68.06%**	55.81%
ProGAN	92.71%	52.24%	**99.86%**
GLIDE	**94.31%**	61.54%	90.50%
Midjourney	74.83%	**86.13%**	76.15%
DALLE-3	66.25%	**91.45%**	41.93%
Imagen 3	90.51%	**98.81%**	89.56%
StyleGAN3	85.94%	49.87%	**90.37%**
Stable Diffusion XL	93.26%	**99.86%**	93.33%

Table 1: Original Detectors' Accuracy Across Different AI Datasets

Dataset	**Original SAFE (ProGAN Train)**	**SAFE Retrained (Version A: ProGAN + Stable Diffusion)**	**SAFE Retrained (Version B: ProGAN + Stable Diffusion + Aligned Forensics)**
Chameleon	55.81%	56.08%	55.99%
ProGAN	99.68%	99.89%	99.92%
GLIDE	90.60%	95.33%	91.32%
Midjourney	76.15%	97.53%	83.49%
DALLE-3	41.39%	46.48%	44.09%
Imagen 3	86.95%	83.81%	93.60%
StyleGAN 3	99.37%	93.19%	93.67%
Stable Diffusion XL	93.33%	97.57%	95.21%

Table 2: Performance of SAFE on AI-Generated Image Datasets After Various Training Data Configurations. Each dataset contains 10k real and 10k fake images.

5 CONCLUSION AND FUTURE WORKS

In this paper, we performed various experiments to compare and attempt to improve the performance of various open-source AI-image detectors to varying degrees of success. We found that trying to change model architectures was both difficult and inefficient, as it would not always guarantee improved performance and the

Method	Chameleon Dataset Accuracy
Base AIDE	62.89%
Base SAFE	55.81%
Aligned Forensics	68.06%
Ensemble (Predicted Label)	59.10%
Ensemble (Confidence Scores)	57.88%
Aligned Forensics Stay-Positive	64.63%

Table 3: Accuracy of Different Methods on Chameleon Dataset. Ensemble takes the predicted label or confidence scores of all three detectors and makes a majority vote decision.

Dataset (Using ZeroFake)	Accuracy
Stable Diffusion XL Base (500 fake)	72.4%
SDXL Base (100 Real)	61%
ProGAN (500 fake)	77.6%
Chameleon (100 fake)	65%
Chameleon (100 real)	26%
SSIM Summation (Chameleon, 70% weight)	61.9%
SSIM Summation (Sigmoid, 70% weight)	61.93%

Table 4: Accuracy of Various Datasets Using ZeroFake

Dataset	Original AIDE	Multi-Scale Patches
Chameleon	62.895%	66.753%
ProGAN	92.713%	91.638%
GLIDE	94.311%	90.522%
Midjourney	74.833%	65.000%
DALLE-3	66.258%	71.020%
Imagen 3	90.511%	85.900%
StyleGAN3	85.944%	85.678%
Stable Diffusion XL	93.267%	84.456%

Table 5: Comparison of Original AIDE and Multi-Scale Patches across Different Datasets

Dataset	Retrained SAFE	Retrained with Mixup
Chameleon	56.08%	56.06%
ProGAN	99.99%	99.79%
GLIDE	95.33%	92.53%
Midjourney	87.35%	84.75%
DALLE-3	46.48%	44.73%
Imagen 3	83.28%	93.29%
StyleGAN3	93.19%	91.76%
Stable Diffusion XL	97.57%	95.62%

Table 6: Performance comparison of retrained SAFE with and without Mixup.

ability to generalize to unseen generation models. We found that focusing on training data selection was an easier and efficient way of improving detector performance. We studied the impact of increasing the size and diversity of a training dataset and found that these methods can improve accuracy and generalization ability of detectors, but can also have diminishing or worse performance when the dataset becomes to large.

We hope that our work highlights the importance of good training dataset selection and curation for training AI-generated image detectors and can inspire future work into examining what makes for high-quality training data along with how much data is optimial for training. Issues in the training process itself should also be examined, such as overfitting and reducing catastrophic forgetting.

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